

Composition of Planet-forming Materials Around Low-Mass Stars with Dana Anderson

Investigating the Dependence of HCO^+ , N_2H^+ and C^{18}O
Column Densities on Stellar and Disk parameters

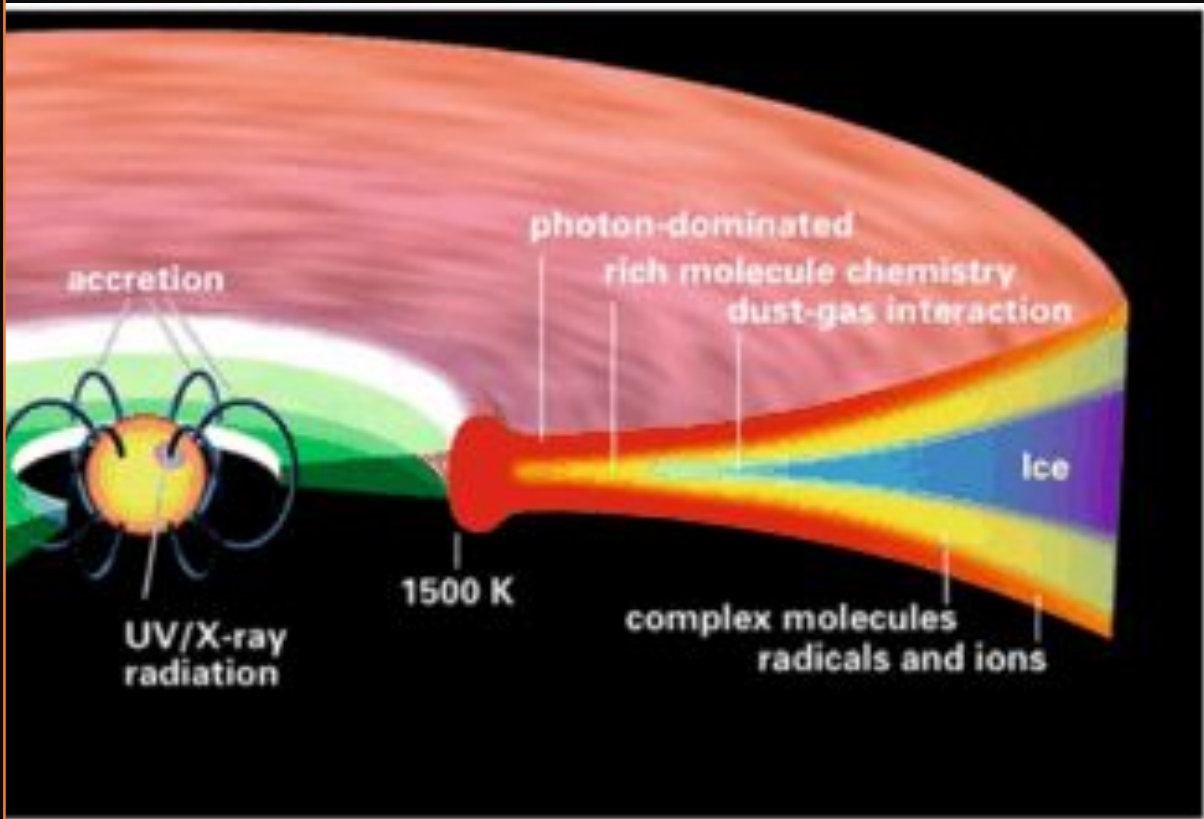
By: Arya Papal

INTRODUCTION

STAR AND PLANET FORMATION

- How are stars and planets formed?
 - What is a protoplanetary disk?
 - Why are we interested?
 - What do they contain?
-

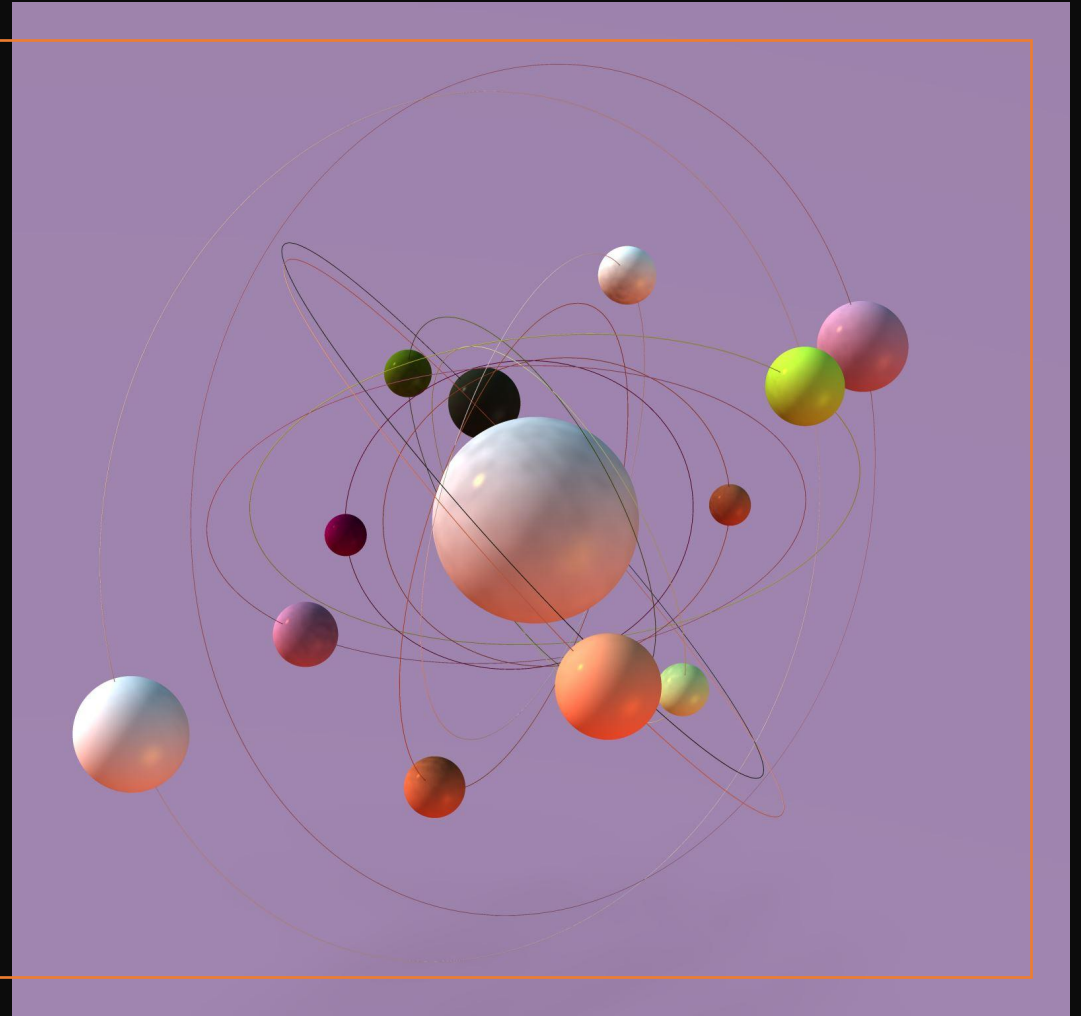
STRUCTURE OF PROTOPLANETARY DISK



- OUTERMOST LAYER: Photon Dominated Layer
- MIDDLE LAYER: Warm Molecular Layer
- INNER LAYER: Frozen Midplane

WHY ARE WE TRACING GASES?

- 200+ molecules have been detected through radio telescopes
- We are specifically looking for HCO^+ , N_2H^+ and C^{18}O gases
- Why?
 - Gas Tracers
 - Impact of Varied Parameters





PARAMETERS

Main source of:

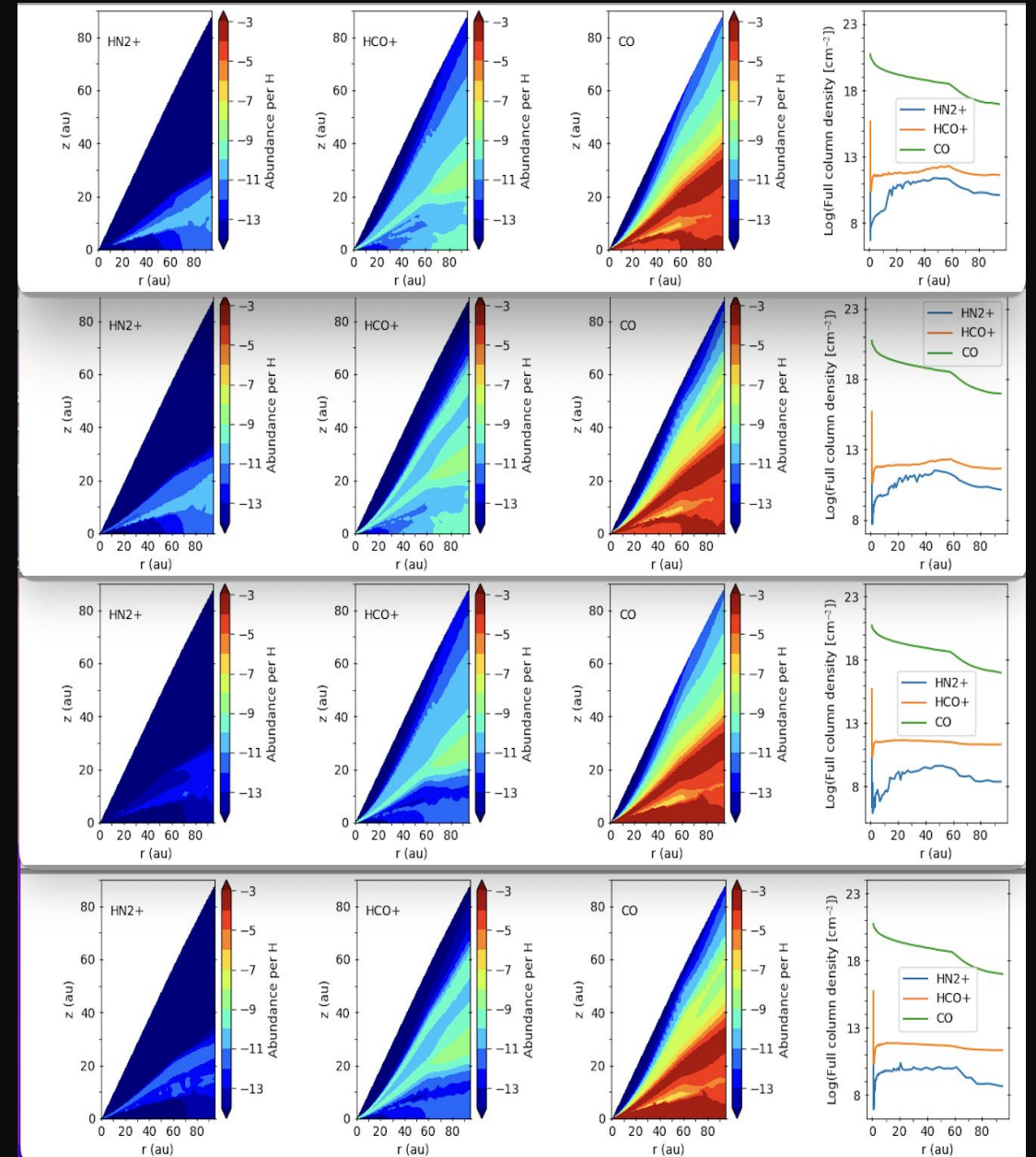
- Temperature: Stellar radiation from K3V, M2V, M5V and M8V stars
- Ionization:
 - Cosmic rays (CR)- high energy particles from background interstellar medium
 - X rays (XR)- from central star

THERMOCHEMICAL MODELING SOFTWARE: rac2d model

This model simulate the chemical Evolution of protoplanetary disk as a function of different Disk parameters and Stellar parameters such as:

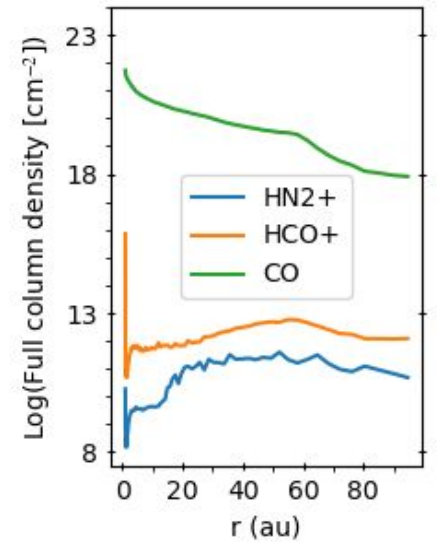
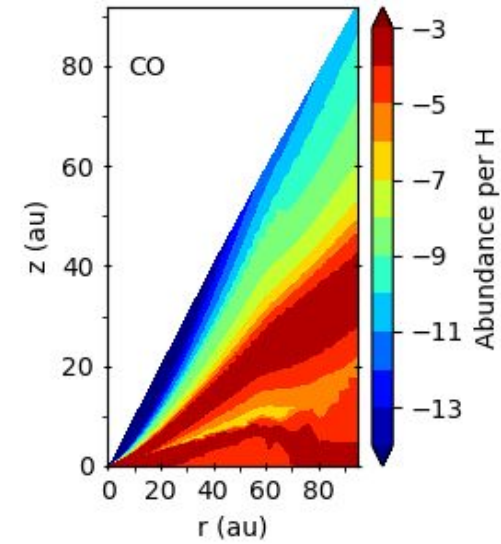
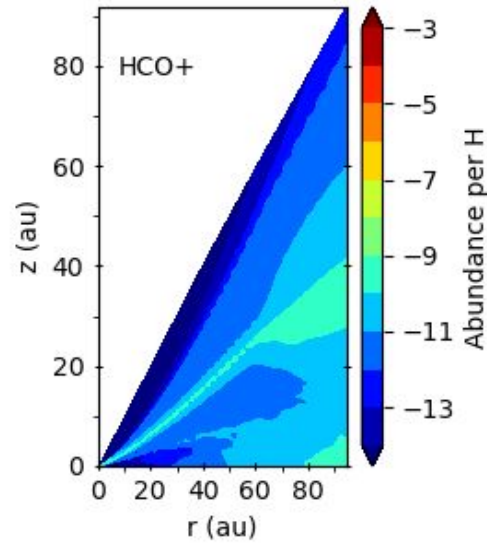
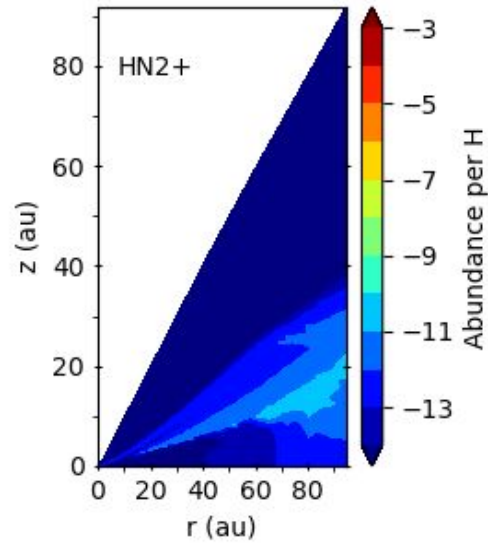
- Disk dust mass
- Gas masses
- UV/X-ray radiation fields
- Disk Sizes
- Central Stellar Mass

We used rac2d to generate chemical disk models for different stars like K3V (0.78 Solar Mass), M2V (0.44 Solar Mass), M5V (0.162 Solar Mass) and M8V (0.085 Solar Mass) for alternating parameters with Low and High Cosmic Radiations (LCR & HCR) with Low and High X-Rays (LXR & HXR).

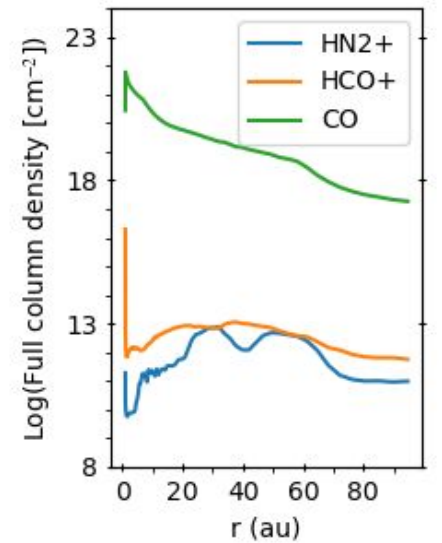
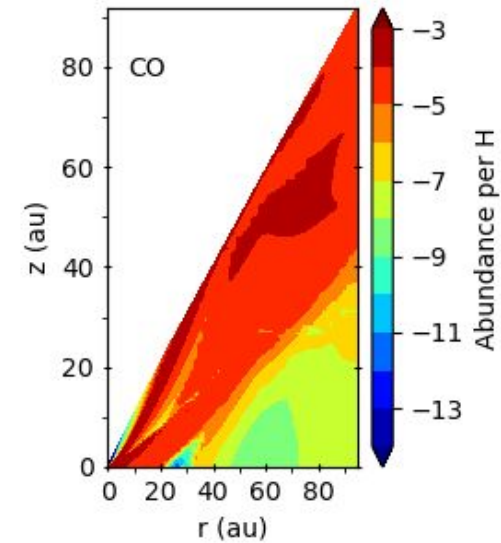
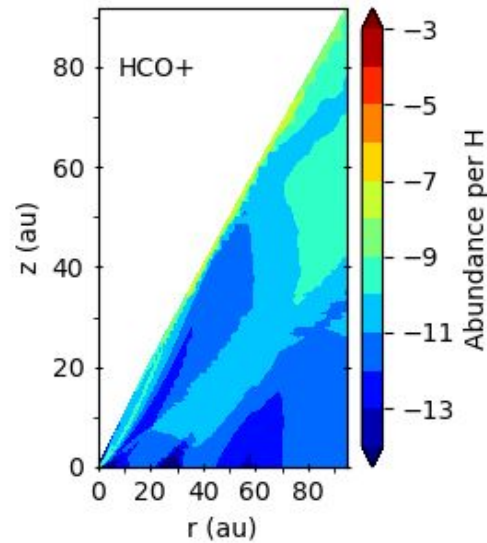
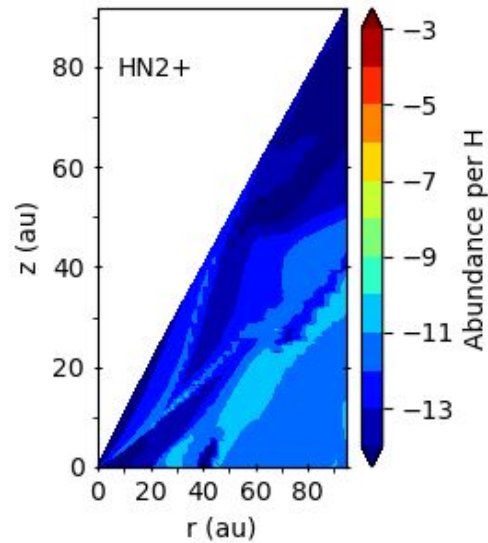


Stellar Mass decreasing (Low XR & Low CR)

K3V
(0.78 Solar Mass)

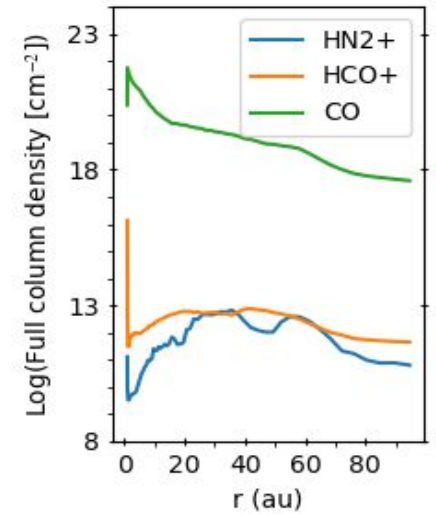
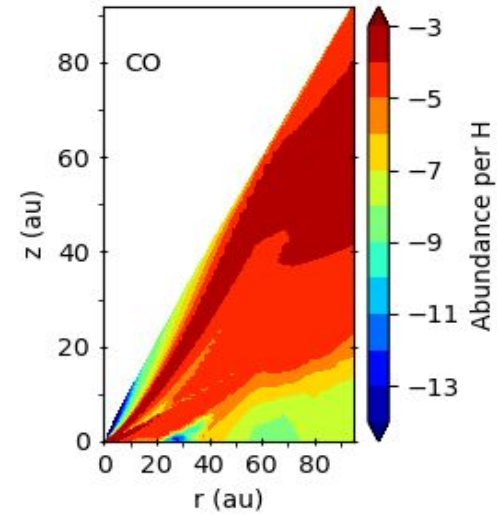
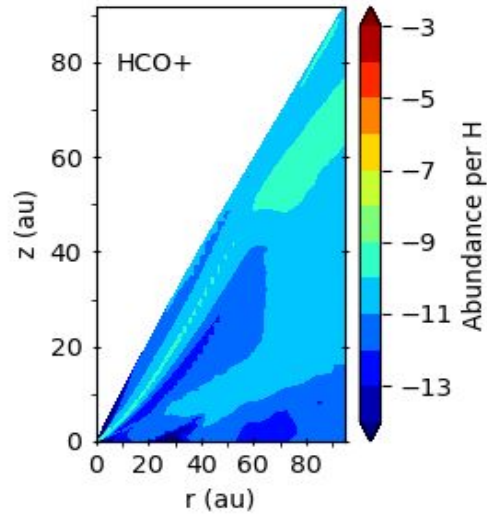
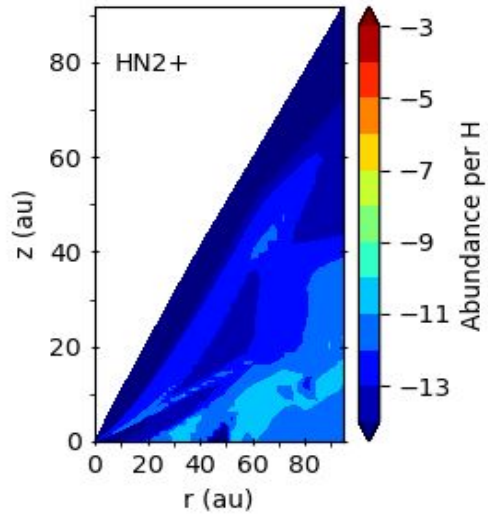


M8V
(0.085 Solar Mass)

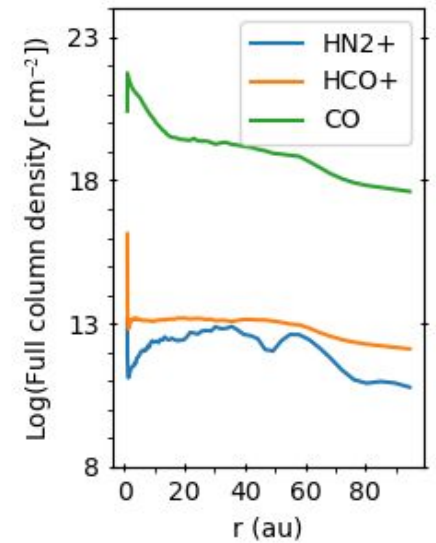
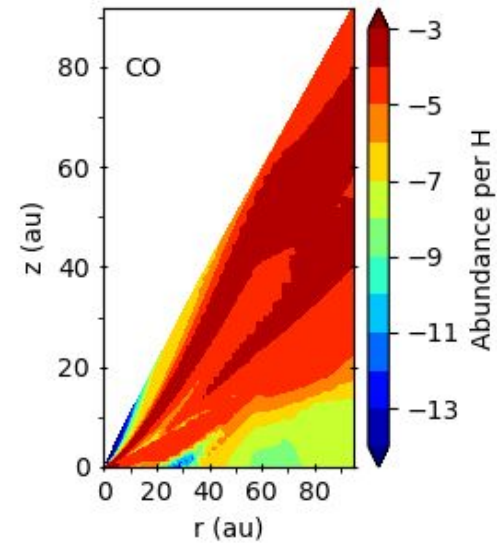
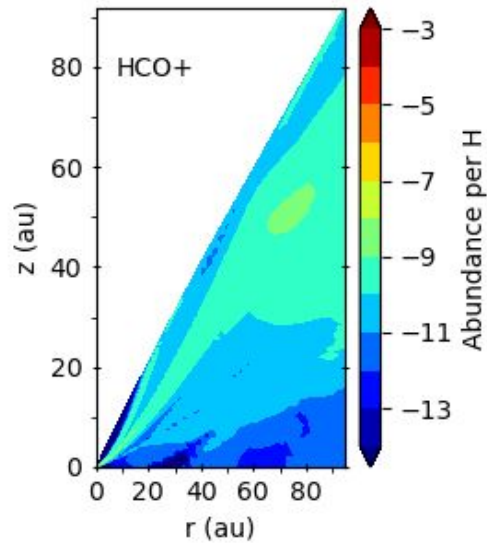
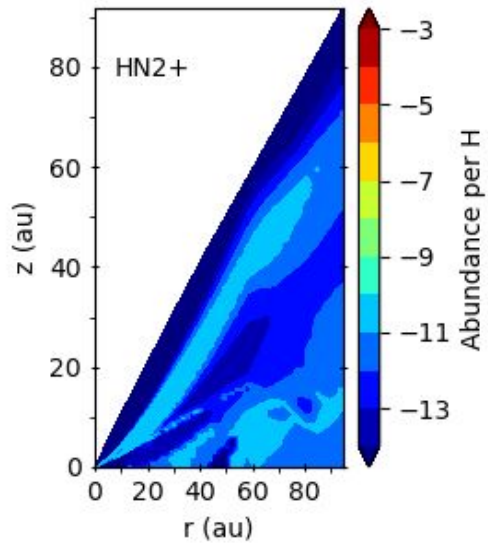


X-ray increasing (M5V- 0.162 Solar Mass)

LCR &
LXR

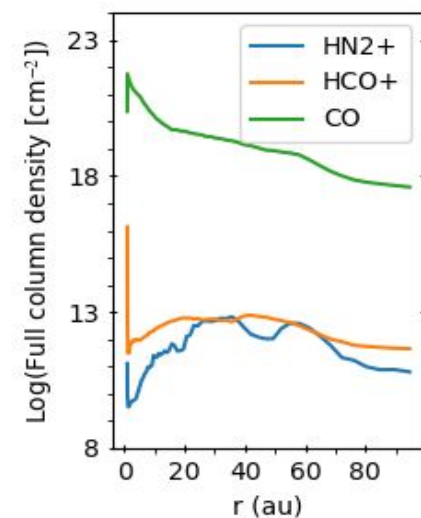
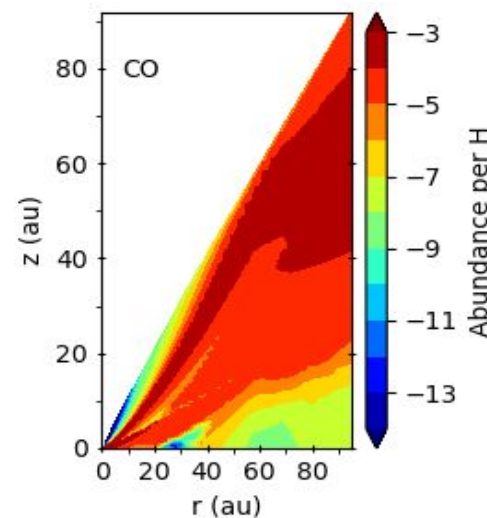
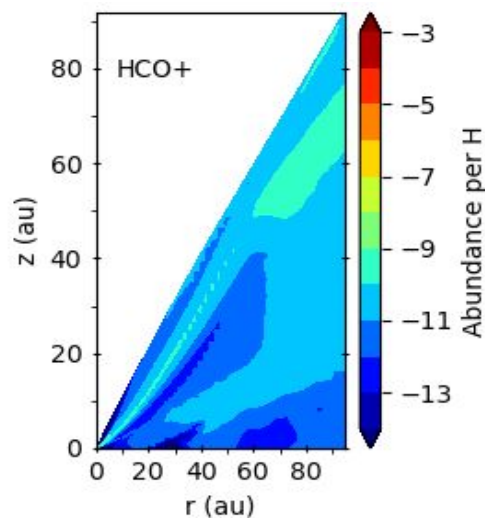
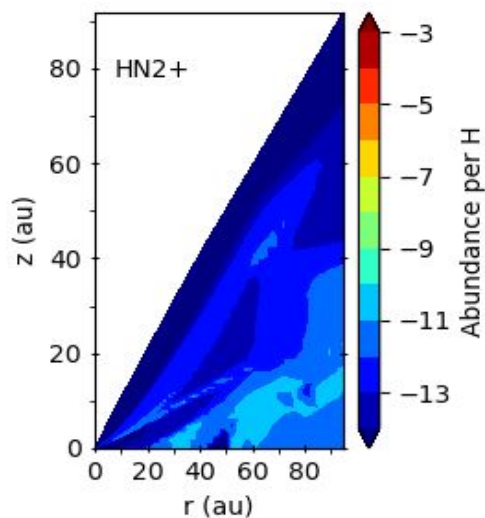


LCR &
HXR

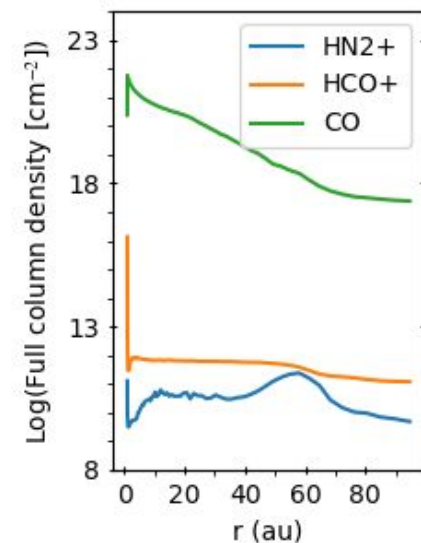
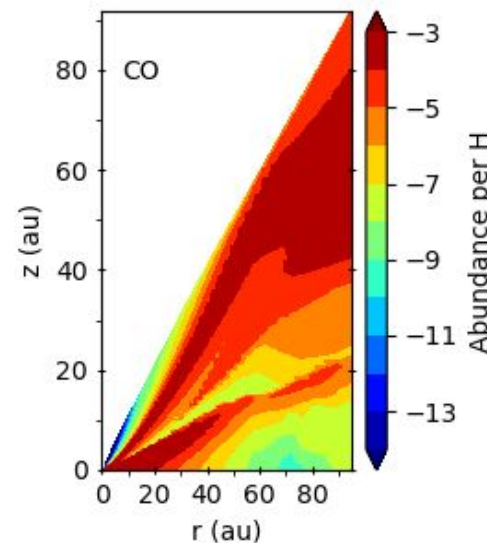
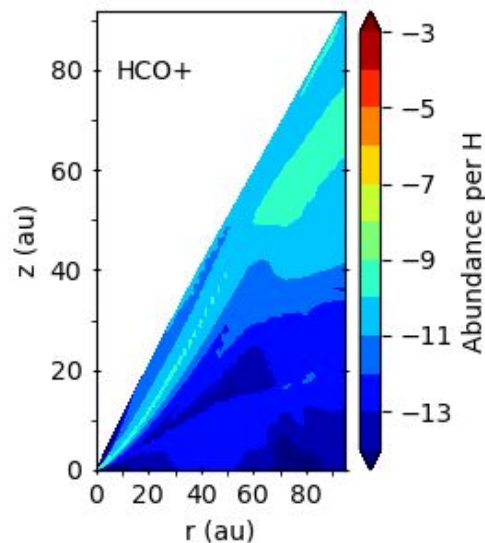
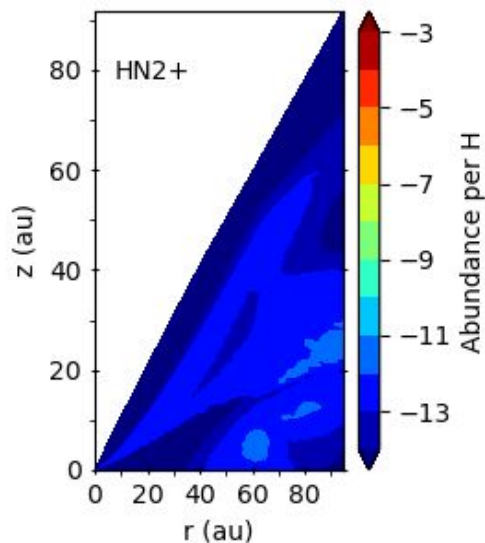


Cosmic Rays increasing (M5V)

LCR &
LXR

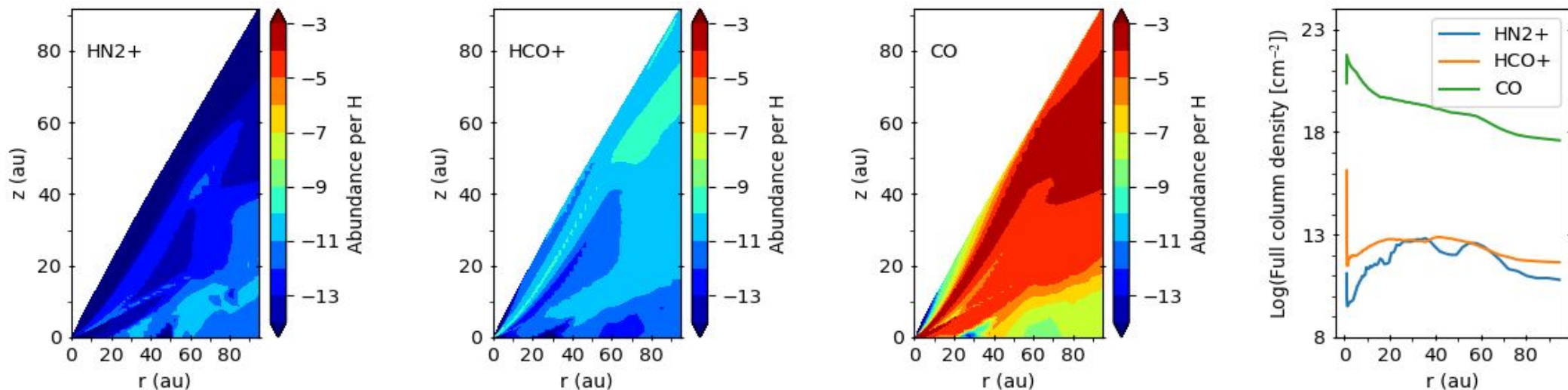


HCR &
LXR

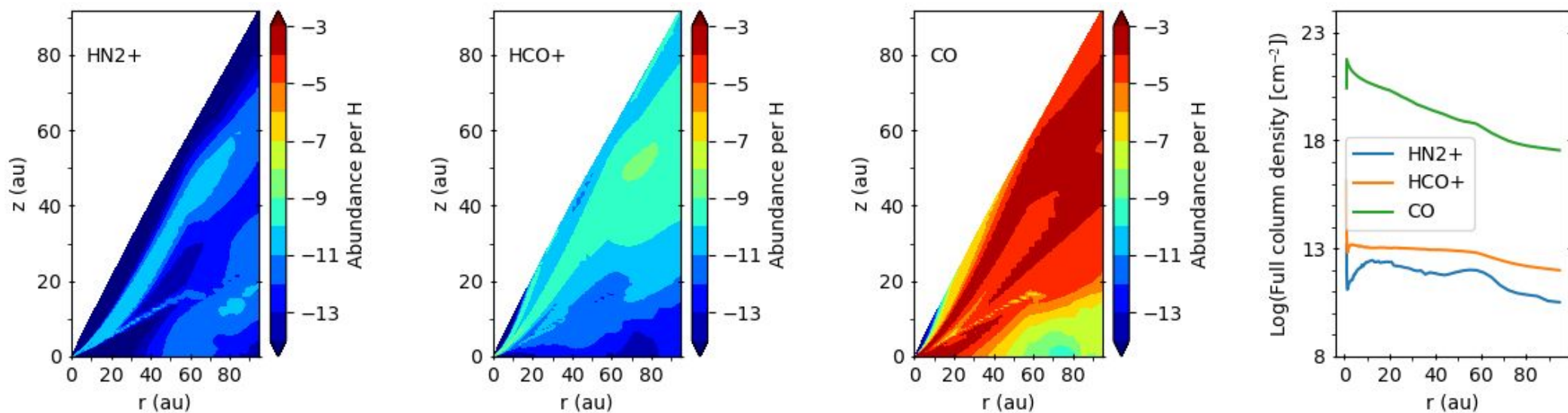


Cosmic Rays and X-Rays increasing (M5V)

LCR &
LXR



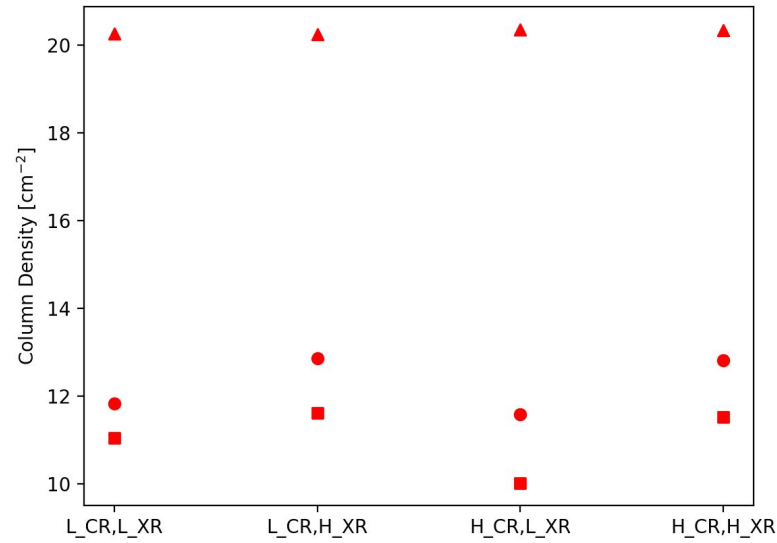
HCR &
HXR



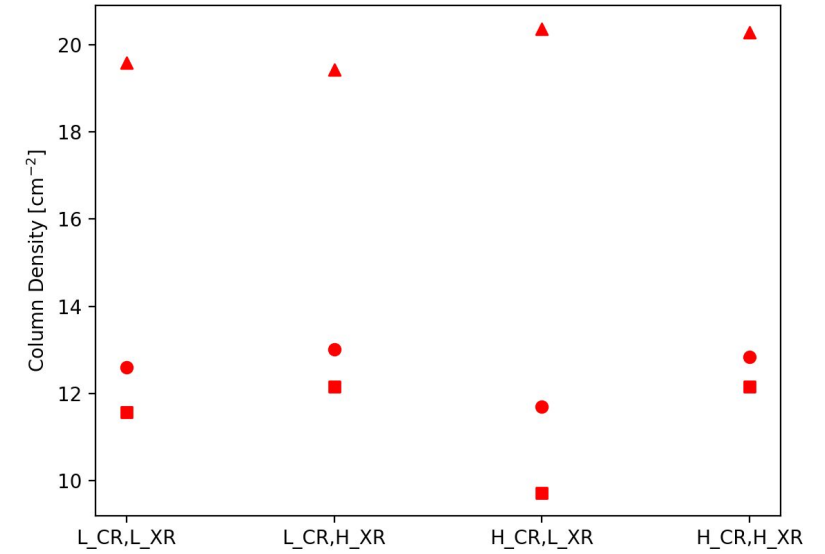
Column Density at radii 20 AU

□ N₂H⁺
○ HCO⁺
△ CO

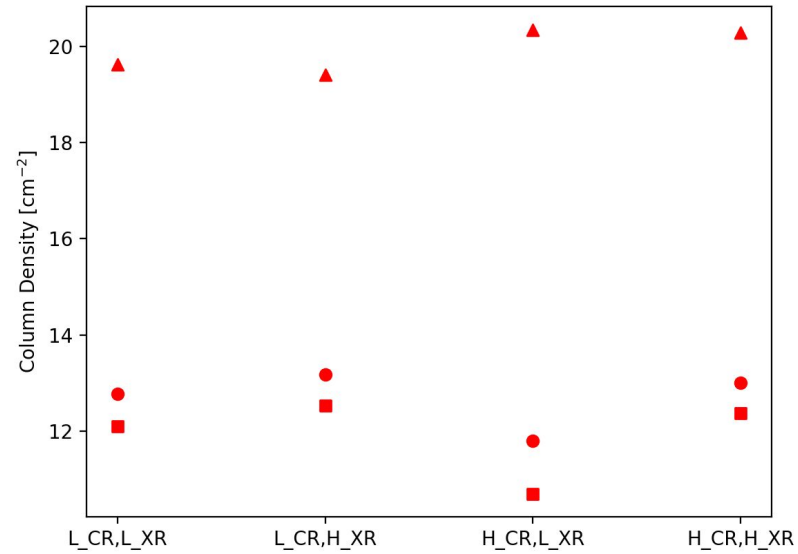
K3V



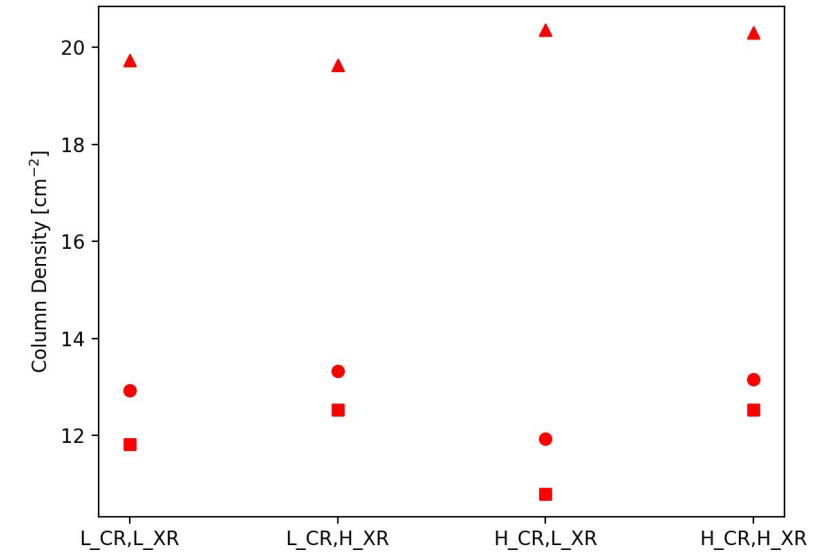
M2V



M5V



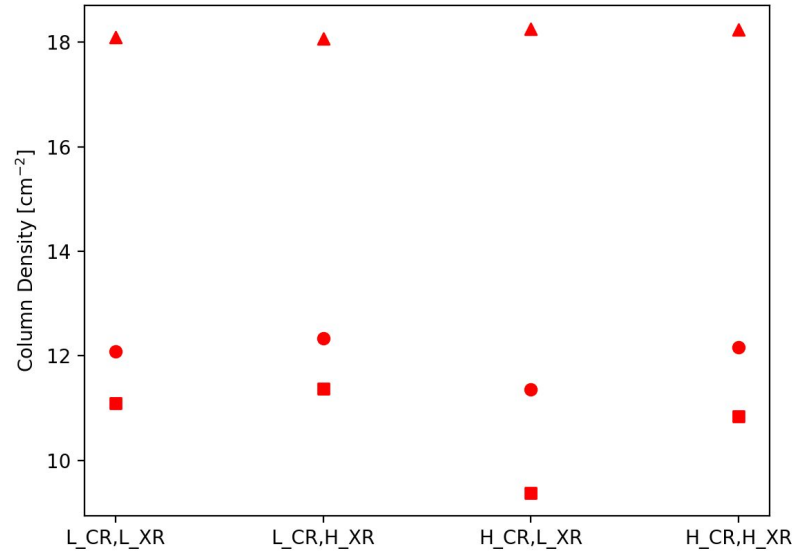
M8V



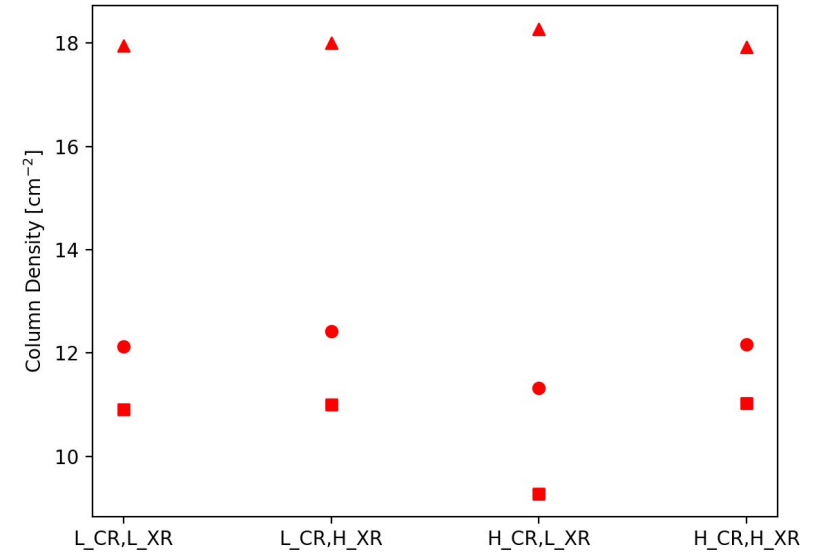
Column Density at radii 80 AU

□ N2H+
○ HCO+
△ CO

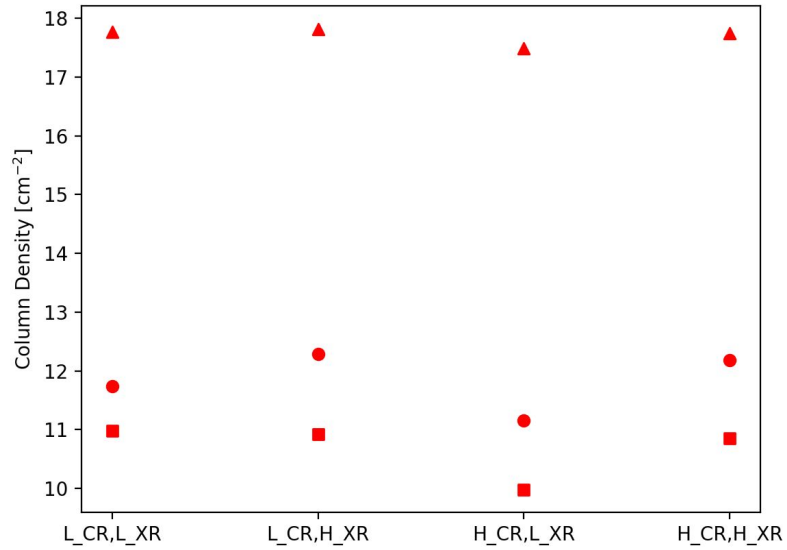
K3V



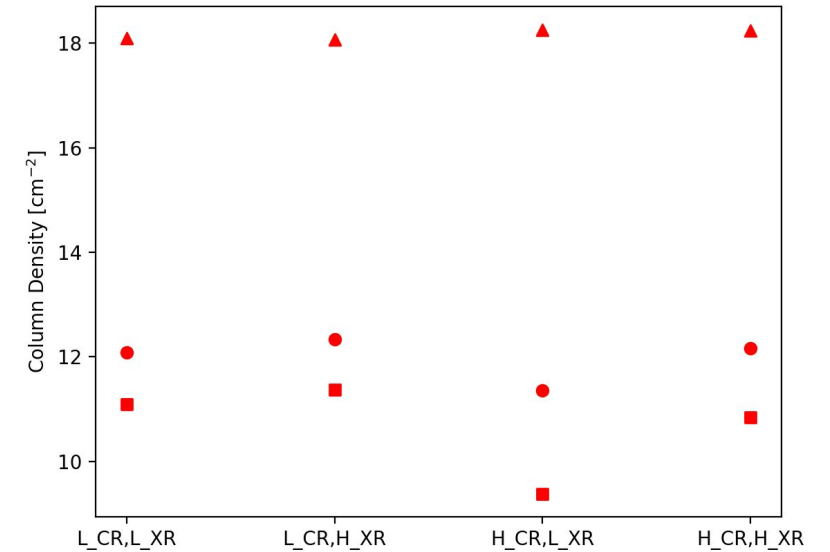
M2V



M5V



M8V



SUMMARY

- We found that HCO^+ , N_2H^+ and C^{18}O serve as effective gas tracers, offering insights into disk evolution and overall structure.
- Our analysis revealed the interplay between different parameters, such as cosmic ray ionization rate ("cr") and X-ray ionization rate ("xr"), in determining the abundances of HCO^+ and N_2H^+ .
- We highlighted the significance of freeze-out for lower stellar masses as a mechanism influencing the observed chemical changes in CO abundance within protoplanetary disks.
- Additionally, our study emphasized the importance of ionization rates in shaping the abundances of key molecular ions.

FUTURE WORK AND NEXT STEPS

- Looking more into fluxes caused by the gases and Observational Validation
- Effects of Cosmic Radiation
- Expanding the model to include Multi-star system
- Increase in CO abundance

TAKE AWAY

- Explored RAC2D model software
- Unveiled astrochemical concepts
- Research Methodology
- Hands-on Learning
- Engaged with fellow researchers



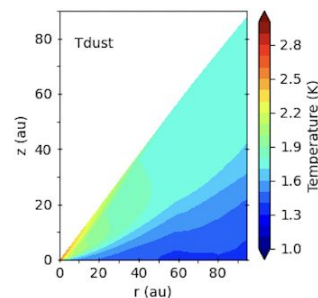
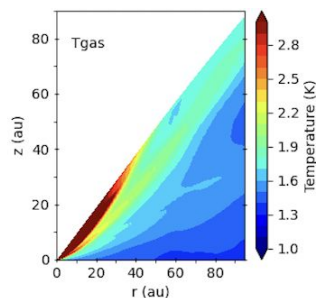
CARNEGIE
SCIENCE

ACKNOWLEDGMENT

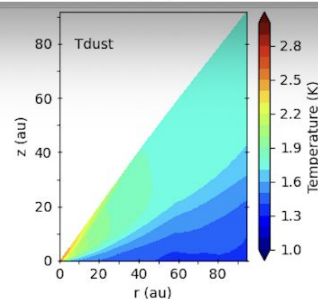
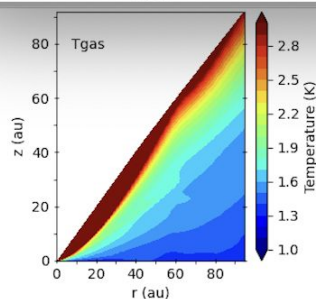
FUTURE WORK AND NEXT STEPS

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- Observational Validation

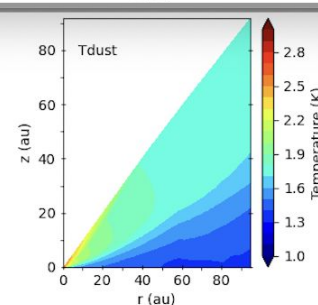
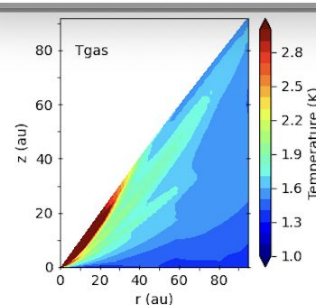
K3V



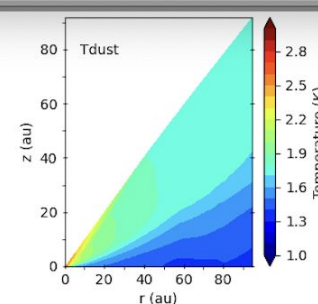
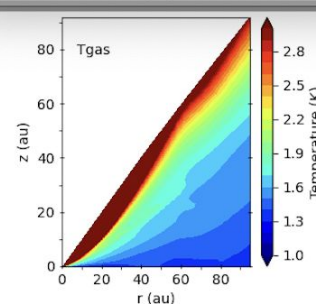
LCR & LXR



LCR & HXR

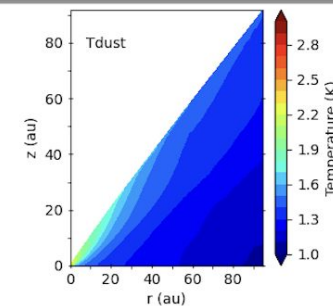
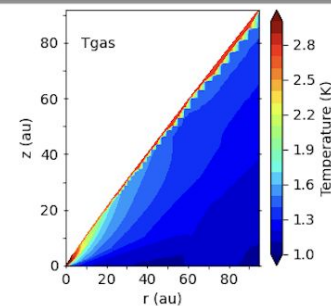


HCR & LXR

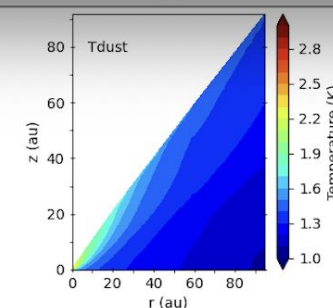
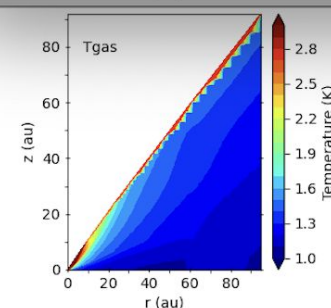


HCR & HXR

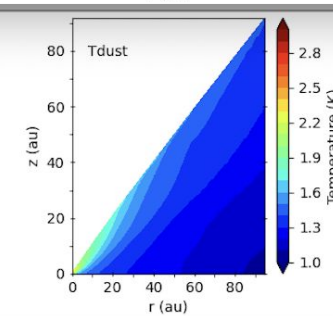
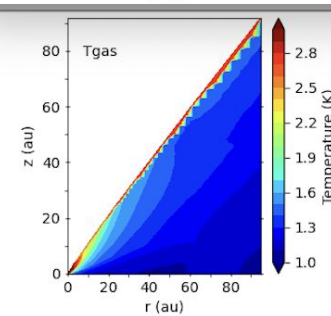
M8V



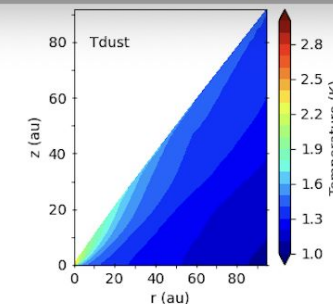
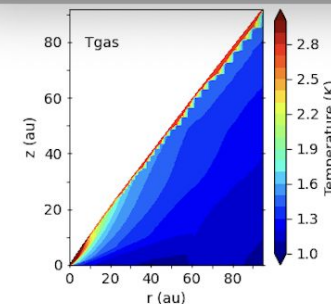
LCR & LXR



LCR & HXR



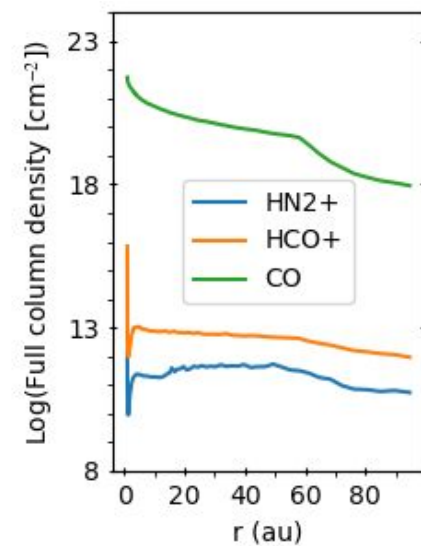
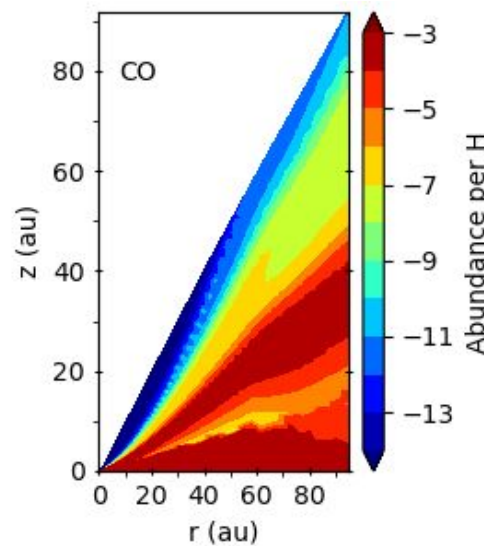
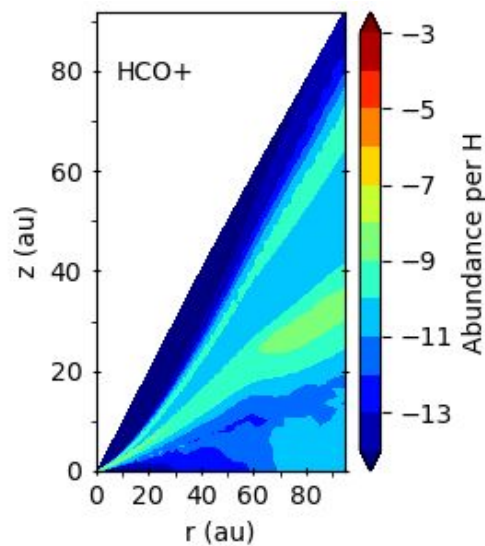
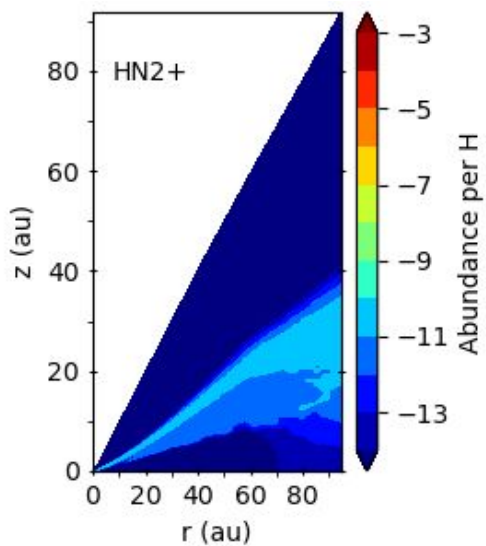
HCR & LXR



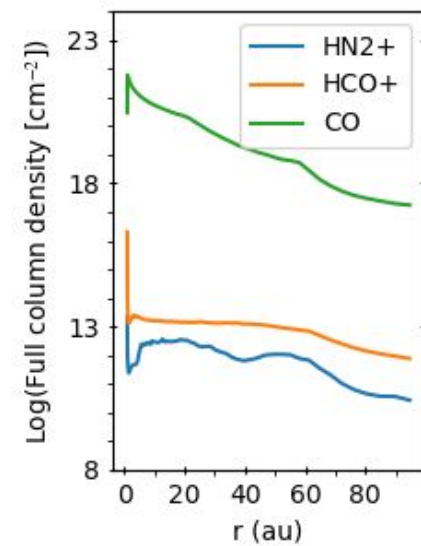
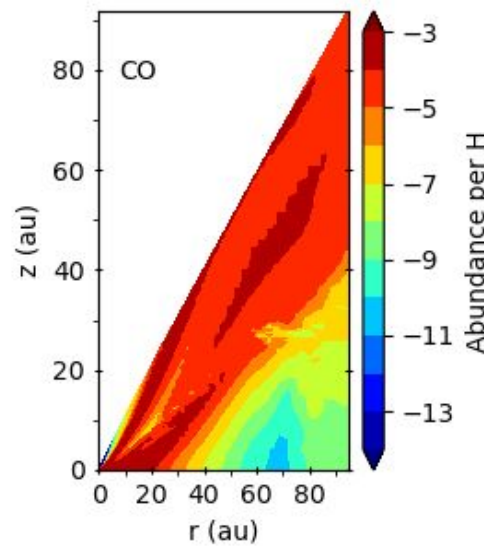
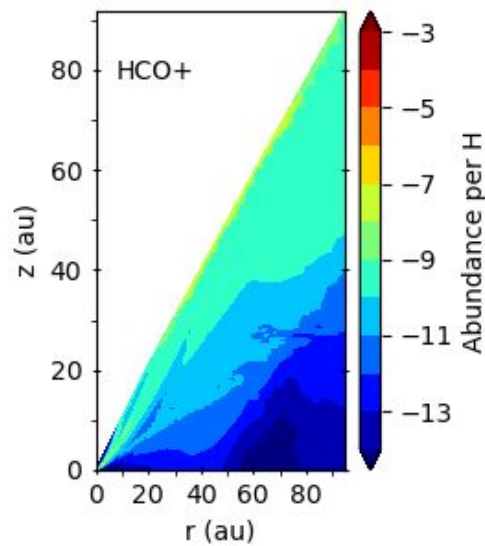
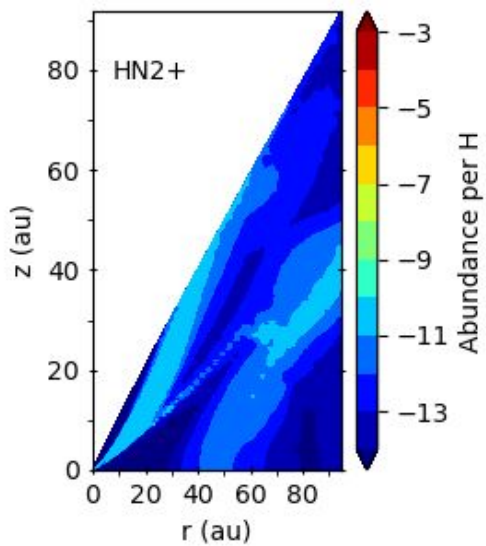
HCR & HXR

Stellar Mass decreasing (High XR & High CR)

K3V

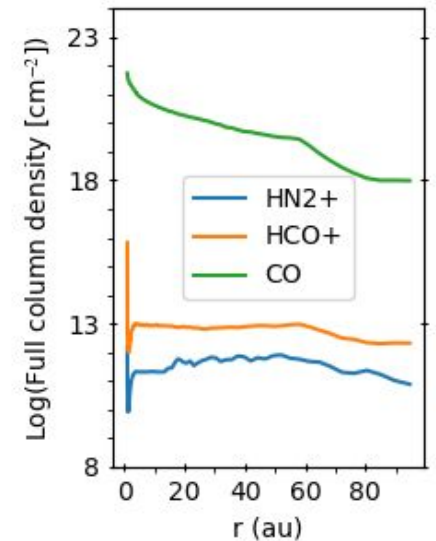
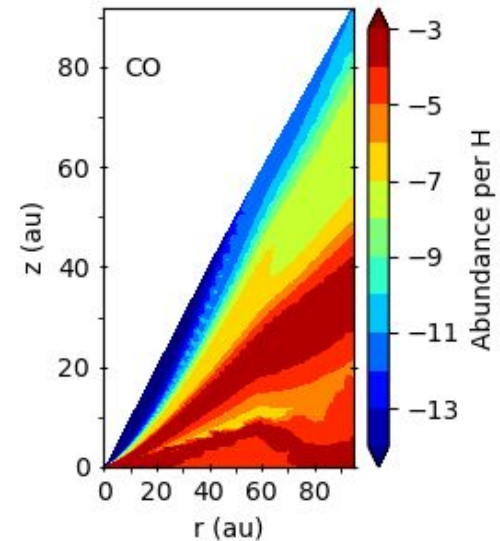
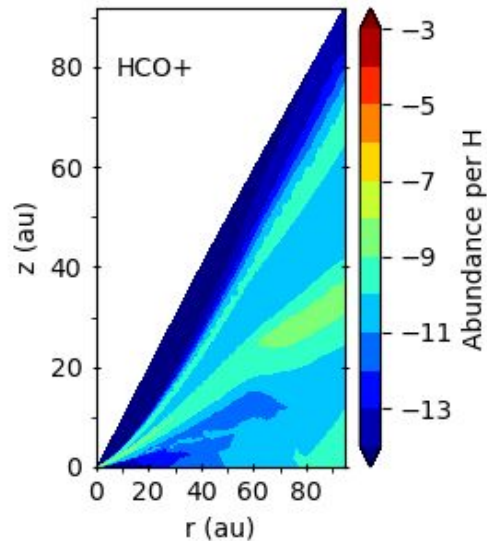
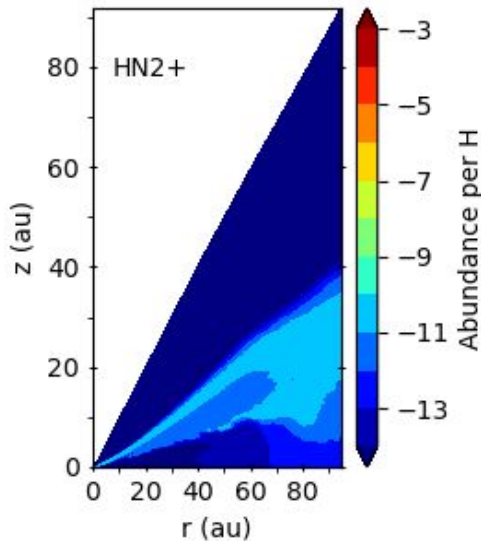


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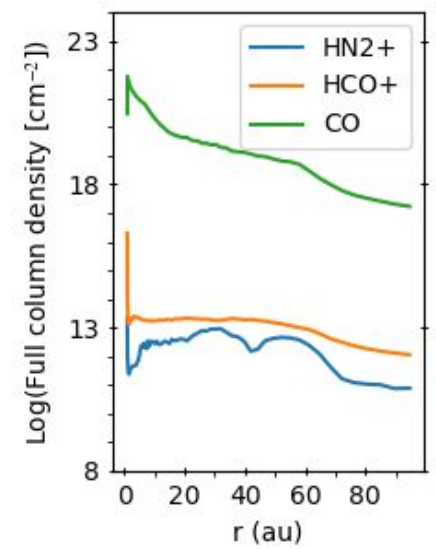
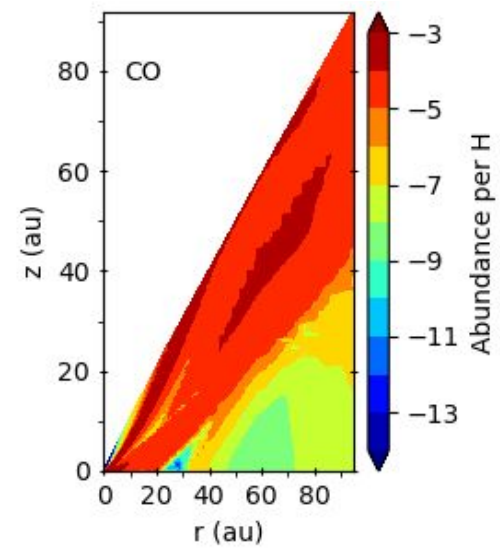
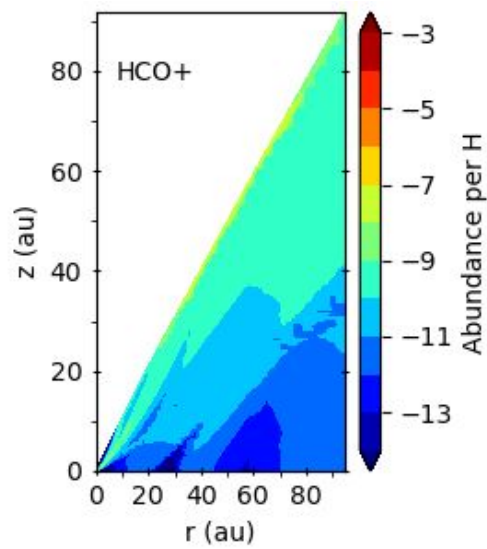
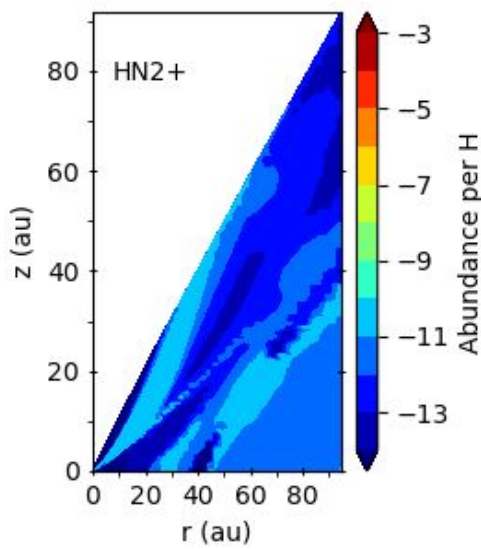


Stellar Mass decreasing (High XR & Low CR)

K3V

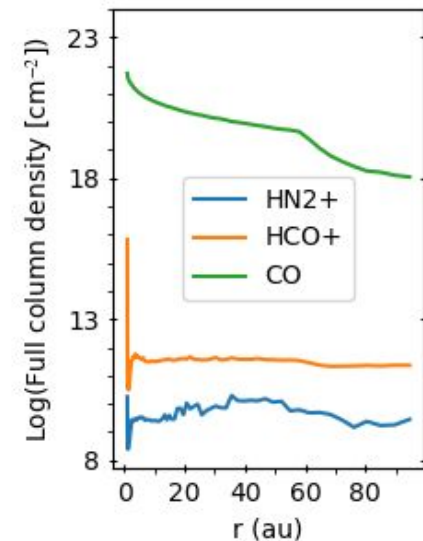
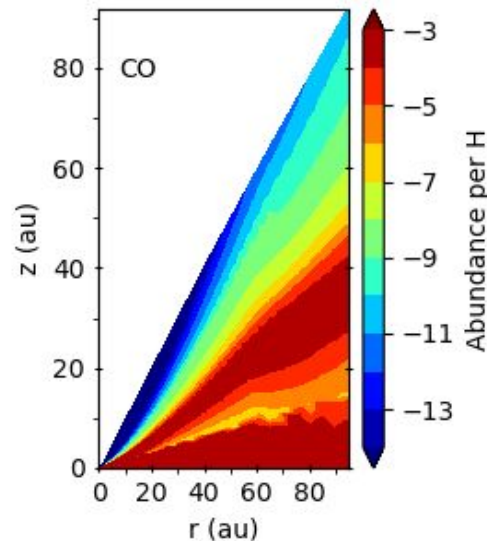
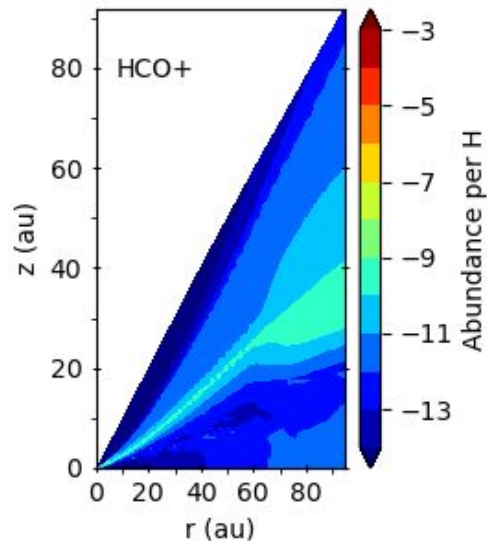
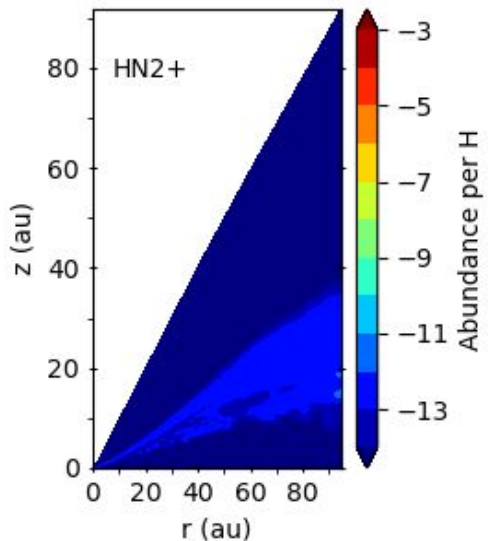


M8V

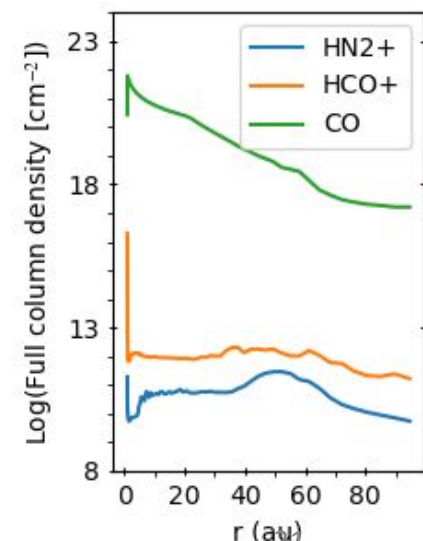
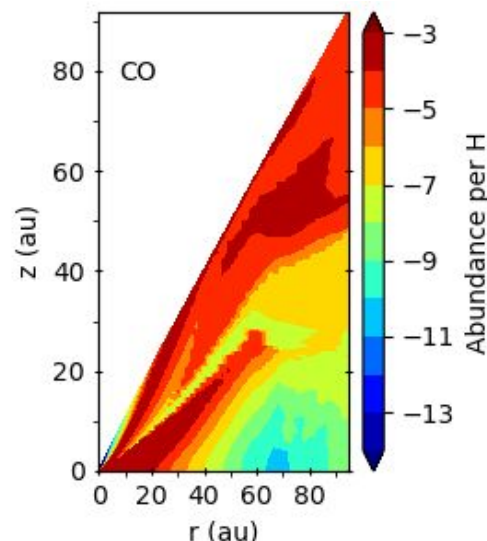
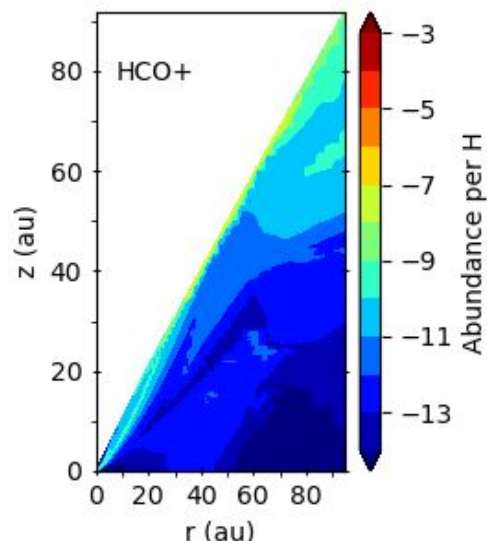
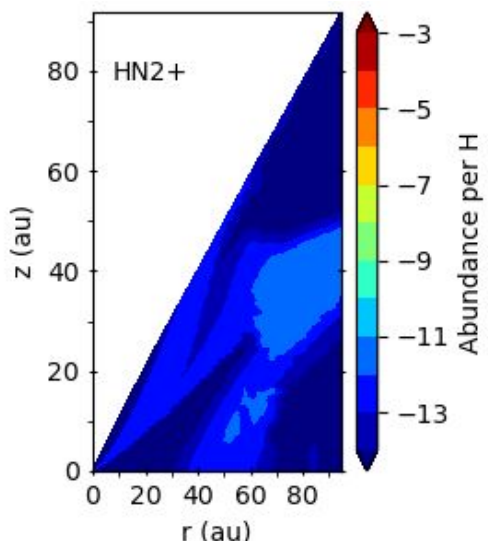


Stellar Mass decreasing (Low XR & High CR)

K3V

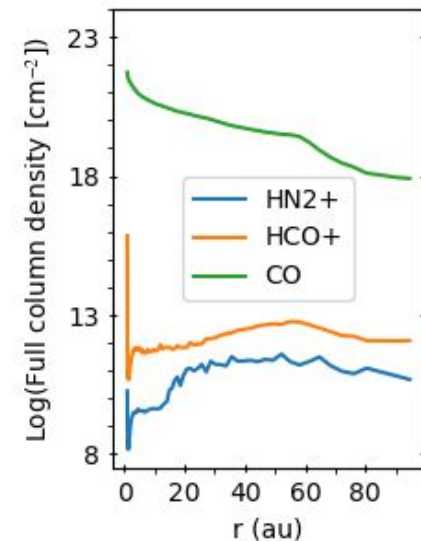
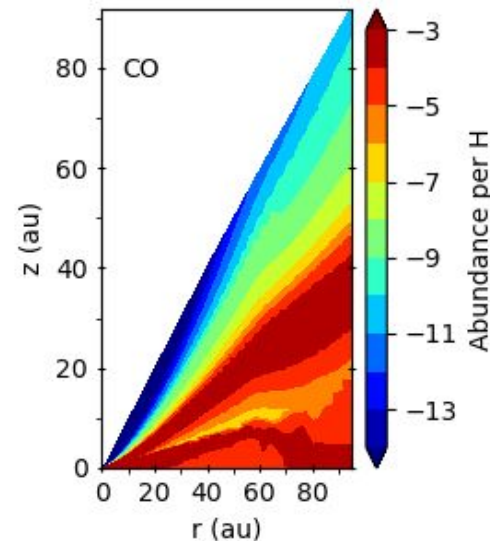
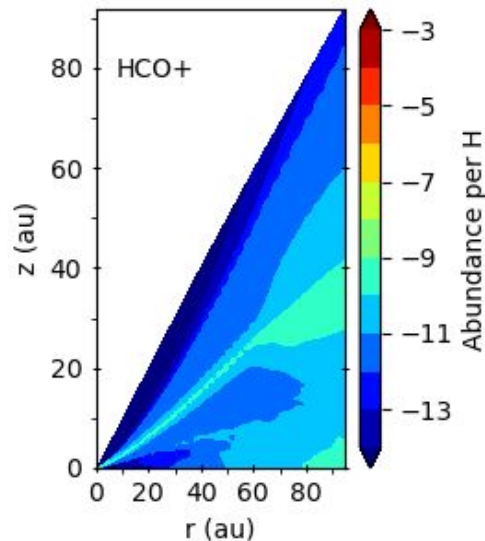
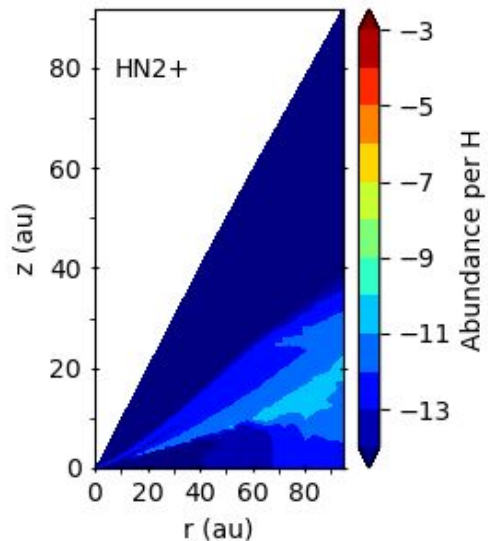


M8V

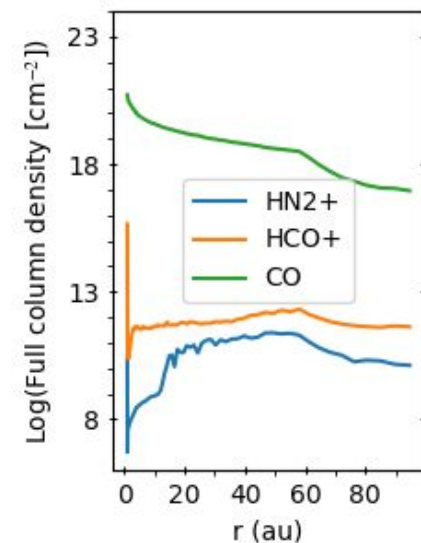
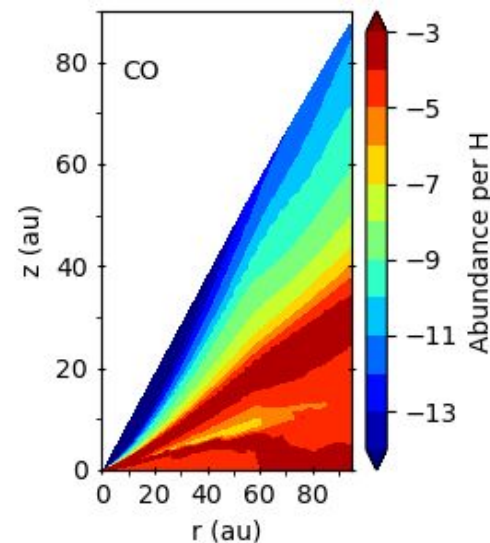
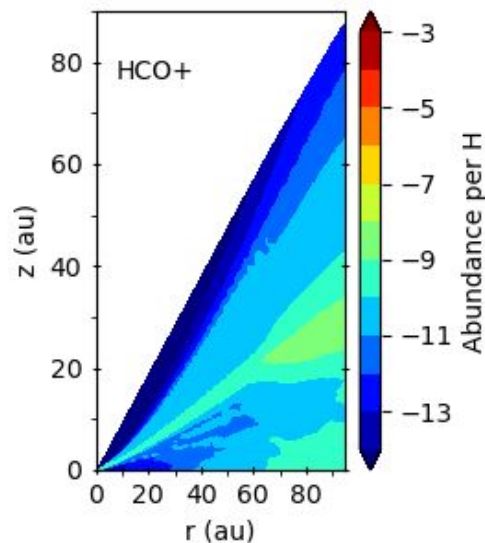
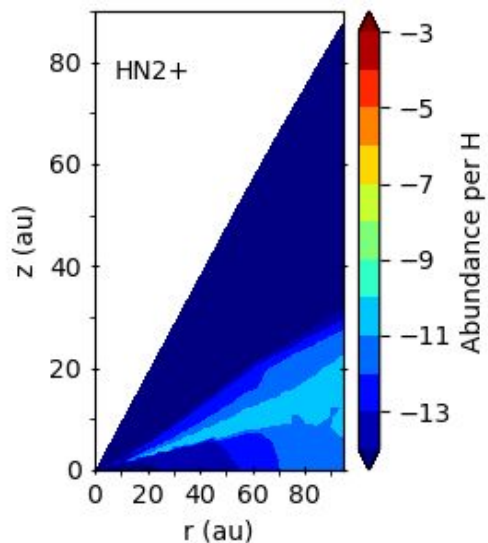


Gas Mass decreasing (Low XR & Low CR)

K3V

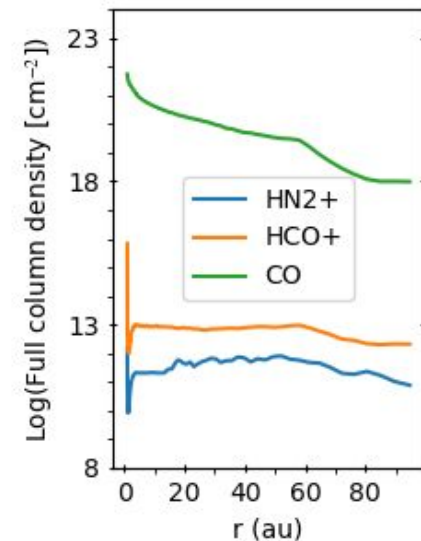
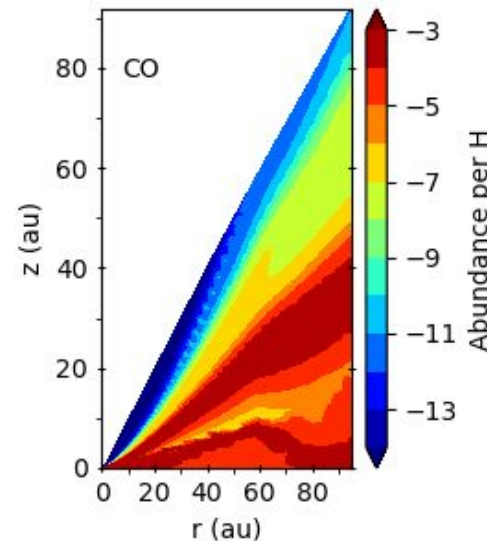
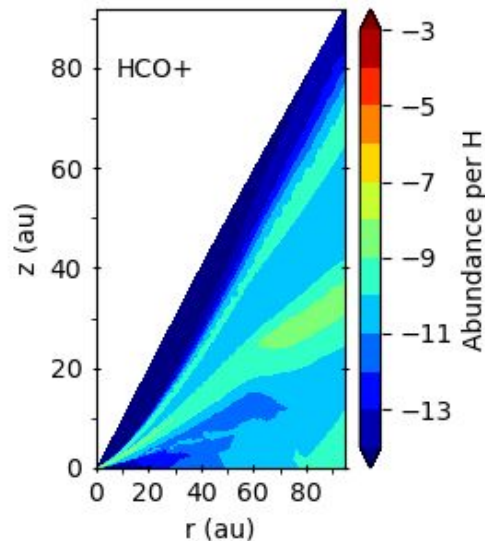
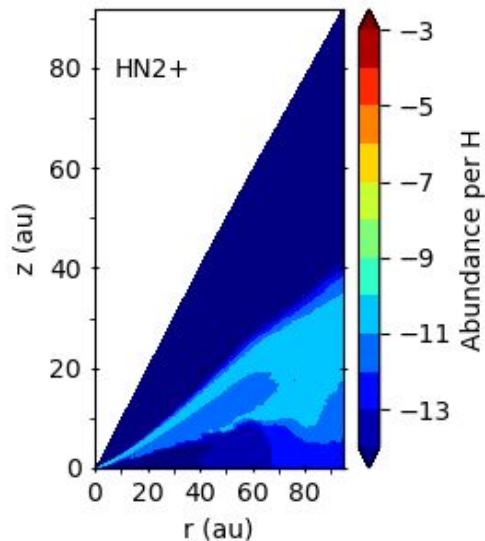


Lower
Gas
Mass

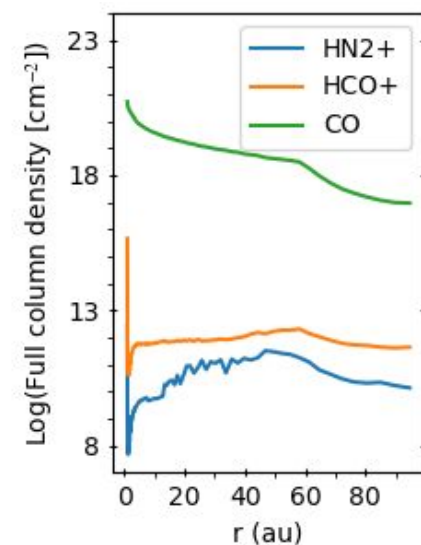
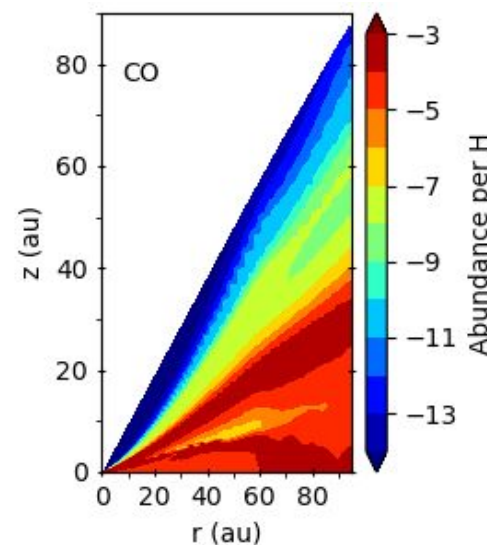
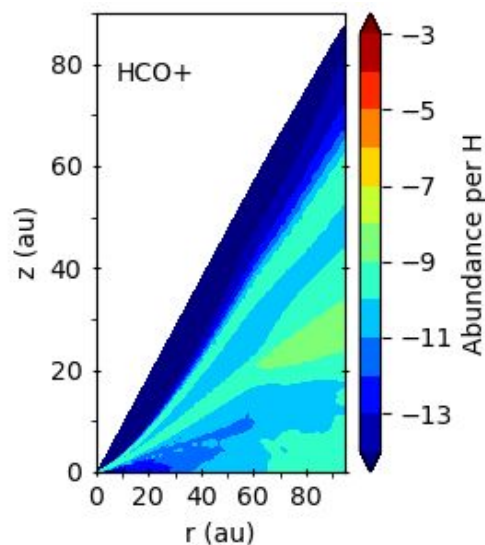
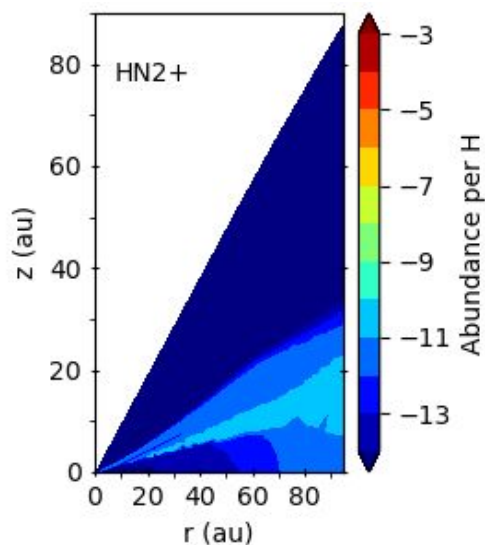


Gas Mass decreasing (High XR & Low CR)

K3V

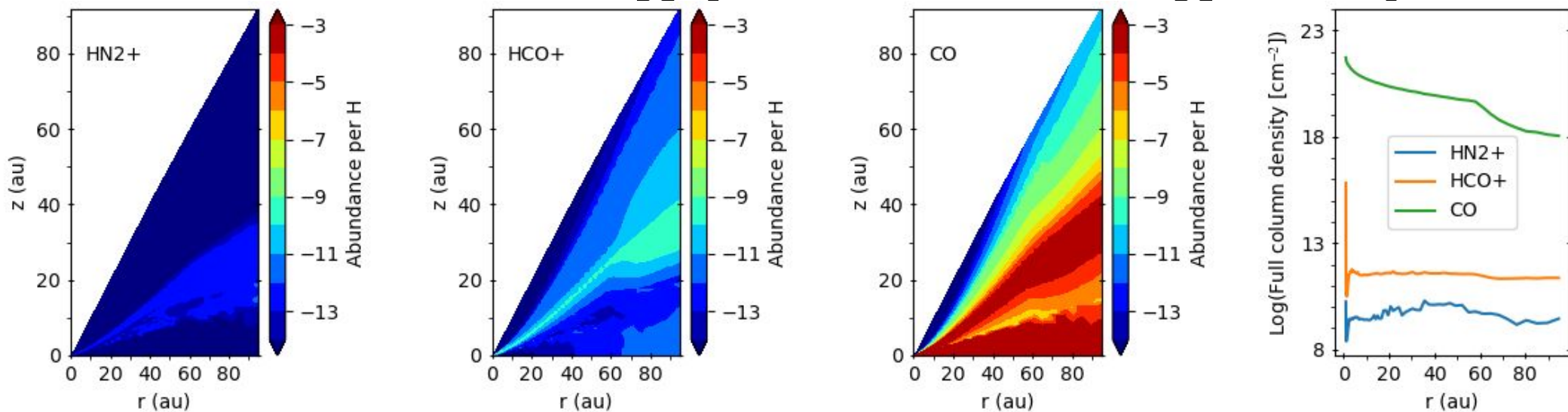


Lower
Gas
Mass

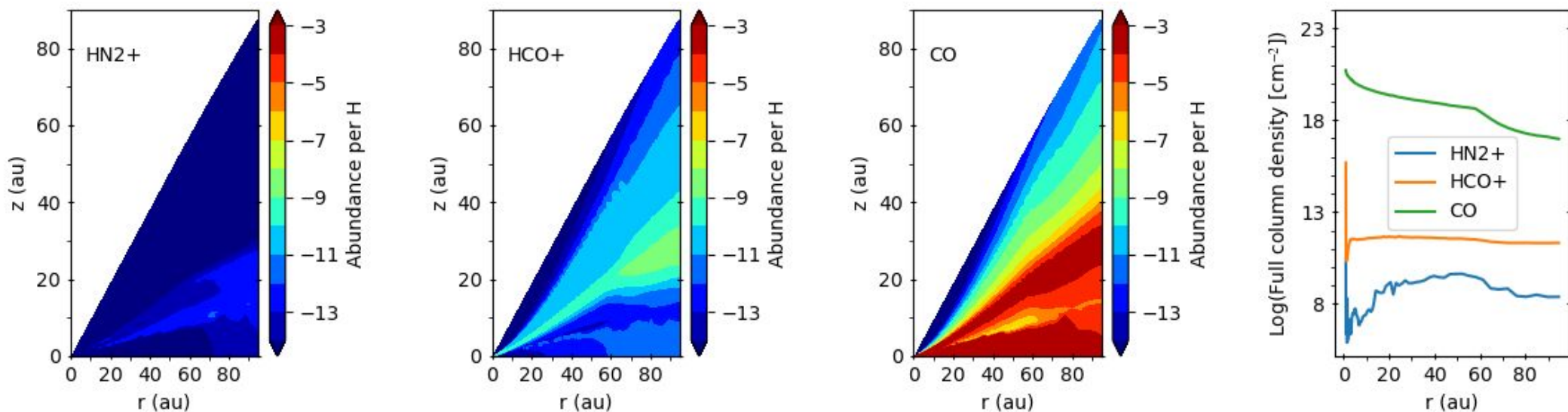


Gas Mass decreasing (Low XR & High CR)

K3V

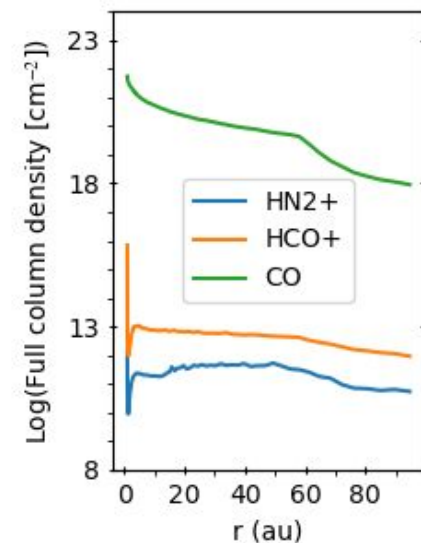
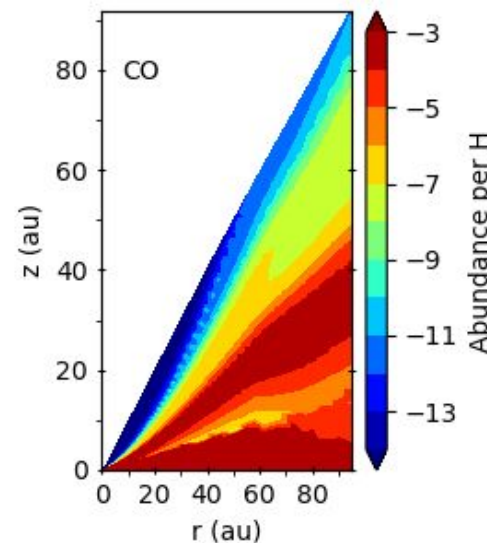
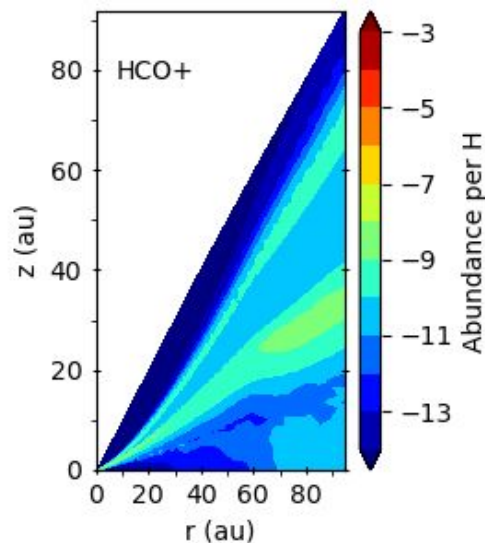
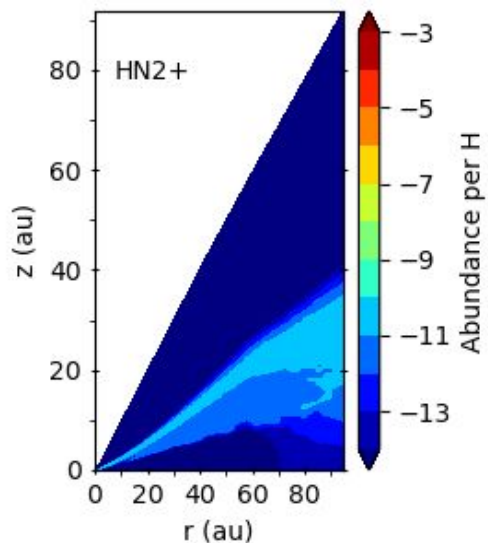


Lower
Gas
Mass



Gas Mass decreasing (High XR & High CR)

K3V



Lower
Gas
Mass

